ZCAS UNIVERSITY

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN SOFTWARE ENGINEERING
BACHELOR OF SCIENCE IN COMPUTING WITH EDUCATION

CIT 4251

STRATEGIC MANAGEMENT OF INFORMATION TECHNOLOGY

FINAL EXAMINATION

TUESDAY 3rd JUNE 2025

TIME: 09.00 – 12.00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS

1. Section A: Question One in Section A is **compulsory**.
2. Sections B: Answer Three (3) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

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SECTION A: QUESTION 1 IS COMPULSORY AND MUST BE ATTEMPTED

Question 1:

Scenario: Read the scenario in Appendix A (also sent to you earlier), and then answer the following questions:

a) Discuss the strategic drivers that SmartTrans pursued to implement a new Logistics Management Information System (LMIS)? (8 Marks)

b) Distinguish the advantages and disadvantages of in-house custom development versus SaaS acquisition for SmartTrans. (8 Marks)

c) Explain the importance of Service-Level Agreements (SLAs) in the SaaS arrangement for SmartTrans. (8 Marks)

d) Discuss the challenges that SmartTrans faced during implementation, and how could these have been mitigated? (8 Marks)

e) If SmartTrans were to expand to Asia, explain the IS considerations that would need to be re-evaluated? (8 Marks)

[TOTAL: 40 Marks]

SECTION B: ATTEMPT ONLY THREE QUESTIONS FROM THIS SECTION

Question 2:

Case Study: Data Breach at HealthSure Clinics – A Wake-Up Call for Strategic IT Security

Background:

HealthSure Clinics is a private healthcare provider with 15 branches across the United Kingdom. It offers general and specialized medical services and stores sensitive patient data in a centralized electronic health records (EHR) system. As part of its digital transformation initiative, HealthSure adopted cloud storage, mobile access for clinicians, and online appointment booking for patients.

In early 2024, HealthSure experienced a **major cybersecurity breach**. An attacker exploited a vulnerability in a third-party mobile app plugin used by doctors for accessing patient records on their tablets. Over 30,000 patient records, including medical histories, billing data, and insurance numbers, were exposed. The incident attracted media attention and regulatory scrutiny under the **UK GDPR**.

Security Posture Before the Breach:

- HealthSure had basic firewall and antivirus protections.
- Multi-factor authentication (MFA) was not enforced on mobile devices.
- No recent penetration testing had been conducted.
- Third-party software was not properly vetted for security.
- The company lacked a formal incident response plan.

Immediate Consequences:

- The Information Commissioner's Office (ICO) launched an investigation.
- Patients lost trust and some transferred to other clinics.
- The company incurred costs for legal support, IT forensics, and breach notification.

- HealthSure faced potential penalties under GDPR for failing to protect patient data.

Post-Breach Measures:

Following the breach, HealthSure's executive team and CIO implemented several strategic improvements:

- Mandatory **MFA** for all system access, including mobile.
- A **formal information security policy** with quarterly risk assessments.
- **Third-party vendor assessments** before integration.
- **Staff training** on phishing and data handling.
- A comprehensive **Incident Response Plan (IRP)** aligned with industry best practices.

The board also approved a **Chief Information Security Officer (CISO)** role to oversee information assurance and regulatory compliance.

Questions:

a) Discuss the strategic implications of lacking an incident response plan for HealthSure. (8 Marks)

b) Describe how employee training contributes to information security in organizations like HealthSure? (8 Marks)

c) Explain the strategic importance of aligning information security with business goals? (4 Marks)

[TOTAL: 20 Marks]

Question 3:

Case Study: FitNation – Leveraging IS for Competitive Advantage in the Fitness Industry

Background:

FitNation is a fast-growing fitness chain in Southeast Asia, offering gym memberships, personal training, and wellness services. Founded in 2018, FitNation gained popularity due to its affordable prices and urban locations. However, by 2023, competition increased with the entry of several tech-enabled fitness startups offering virtual workouts, health tracking apps, and AI-based coaching.

Recognizing the threat, FitNation's leadership team decided to use **information systems strategically** to maintain and grow their competitive advantage.

Strategic Use of IS:

FitNation invested in a digital transformation program with three main components:

1. **Member Experience App:** A mobile application that allows users to:
    - Book classes and personal trainers.
    - Track workouts and nutrition.
    - Access virtual training sessions.
2. **Data Analytics Platform:** An integrated system that collects and analyzes member data to:
    - Personalize training programs.
    - Recommend health products and services.
    - Predict membership churn and improve retention.
3. **IoT-Enabled Equipment:** Smart gym equipment that tracks usage, detects maintenance needs, and syncs with members' fitness profiles in real time.

Outcomes and Competitive Advantages:

By 2024, FitNation reported:

- A 35% increase in member engagement due to personalized services.
- A 22% drop in churn through predictive analytics.
- Revenue growth from upselling health supplements via the app.
- Higher equipment uptime and reduced maintenance costs.

FitNation positioned itself as a **tech-enabled fitness leader**, differentiating its brand and increasing customer loyalty.

Questions:

a) Discuss how FitNation used IT to respond to Porter's five forces, particularly the threat of new entrants. (8 Marks)

b) Identify and explain one example of a value chain activity enhanced by FitNation's information systems and give its impact. (8 Marks)

c) Explain the potential risks of FitNation's reliance on data-driven decision-making? (4 Marks)

[TOTAL: 20 Marks]

Question 4:

Case Study: CollabTech Ltd – Using Social Media for Internal Collaboration and Customer Engagement

Background:

CollabTech Ltd is a mid-sized software development firm based in Dublin, Ireland. With over 400 employees across Europe and North America, the company specializes in enterprise productivity solutions. As CollabTech grew, it began experiencing challenges in internal communication, cross-functional collaboration, and customer support consistency.

To address these issues, the Chief Information Officer (CIO) initiated a strategy to leverage **enterprise social media tools** to enhance both **internal collaboration** and **external engagement**.

The Social Media Strategy:

1. **Internal Collaboration – Enterprise Social Network (ESN):**
    - Deployed **Microsoft Yammer** for internal communication.
    - Created topic-based groups for departments, projects, and initiatives.
    - Used polls, knowledge sharing, and recognition features to promote engagement.
2. **Customer Support and Feedback – External Social Media:**
    - Monitored **Twitter and LinkedIn** for customer queries and issues.
    - Launched a **branded online community** where users share tips, report bugs, and suggest features.
    - Integrated chatbots and AI tools to respond to FAQs on Facebook Messenger.
3. **Marketing and Thought Leadership:**
    - Used **LinkedIn and YouTube** to share webinars, blog posts, and product demos.

- Encouraged employees to share corporate content through their personal networks.

Outcomes:

- Internal response time improved by 30%.
- Project team collaboration across countries increased by 45%.
- Customer satisfaction scores increased due to faster social media responses.
- A new product idea originated from the user community and was adopted into development.
- Employee engagement rose through open recognition on the ESN.

Questions:

a) Discuss two strategic benefits of enterprise social networking tools like Yammer for CollabTech? (8 Marks)

b) From a strategic perspective, explain the importance of social listening for CollabTech? (8 Marks)

c) Give two potential risks associated with CollabTech's use of social media, and describe how they can be mitigated? (8 Marks)

[TOTAL: 20 Marks]

Question 5:

Case Study: StyleNest – Growing a Fashion E-Commerce Brand through Strategic B2C Technology

Background:

StyleNest is a Nigeria-based online fashion retailer targeting urban millennials across West Africa. Launched in 2021, it offers trendy clothing, accessories, and beauty products through its e-commerce website and mobile app. By mid-2023, StyleNest had attracted a large social media following but struggled with cart abandonment, delivery delays, and low customer retention.

The management team realized that to thrive in the competitive B2C space, they needed to strategically leverage e-commerce technologies and business models to improve customer experience and operational efficiency.

Strategic Initiatives:

In early 2024, StyleNest implemented several e-commerce strategies:

1. **Improved Website Design & Mobile Optimization:**
    - Faster loading pages.
    - Personalized product recommendations.
    - Mobile-first layout for better user experience.
2. **Multiple Payment Gateways:**
    - Credit/debit cards, mobile money, and "Pay on Delivery" options.
3. **Customer Relationship Management (CRM) System:**
    - Tracks customer purchases, preferences, and complaints.
    - Enables personalized email campaigns and loyalty rewards.
4. **AI-powered Chatbot:**
    - Provides 24/7 support for order tracking, FAQs, and product queries.
5. **Third-Party Logistics Integration:**
    - Real-time tracking and better last-mile delivery.

Outcomes:

By early 2025, StyleNest achieved:

- 38% increase in mobile sales.
- 30% reduction in cart abandonment.
- 22% improvement in customer retention rate.
- Positive customer feedback on faster delivery and responsive support.

StyleNest continues to scale regionally and considers expanding into influencer-based live shopping experiences.

Questions:

a) Identify the type of B2C e-commerce model that StyleNest is using, and explain how it creates value for the business? (8 Marks)

b) Describe how the AI-powered chatbot supports both customer service and business goals? (8 Marks)

c) Explain how StyleNest can use social media to enhance its B2C strategy further? (4 Marks)

[TOTAL: 20 Marks]

APPENDIX A

Case Study: SmartTrans Logistics – Navigating IS Development and Acquisition Decisions

Background:

SmartTrans Logistics is a mid-sized logistics company based in Cape Town, South Africa, with operations in several African countries. Established in 2008, SmartTrans offers shipping, freight forwarding, and supply chain solutions to small and medium-sized enterprises (SMEs). With increasing competition and evolving customer expectations, the company recognized that its legacy information system—used to manage orders, track shipments, and handle inventory—was no longer meeting performance or scalability requirements.

In 2023, the board approved an initiative to develop or acquire a new integrated logistics management information system (LMIS) to enhance operational efficiency, customer experience, and data-driven decision-making.

Strategic IS Needs:

The proposed LMIS would support the following strategic objectives:

- Real-time tracking of shipments via GPS and RFID.
- Enhanced analytics for route optimization and cost reduction.
- Integration with third-party logistics (3PL) providers and customer ERP systems.
- Mobile access for drivers and warehouse staff.
- Compliance with European transport regulations and data privacy laws.

The CIO, Marta De Vries, formed a cross-functional team to evaluate whether the LMIS should be built in-house, acquired as off-the-shelf software, or outsourced to a third-party developer.

The Decision Process:

The team performed a feasibility study, including cost-benefit analysis and risk assessment. They identified five main options:

1. **Custom Development In-House:** Building the system with the internal IT team.
2. **Open Source Logistics Platform:** Customizing an existing open-source system.
3. **Off-the-Shelf Software (Commercial):** Purchasing an established logistics system.
4. **Outsourcing to a Software Development Firm:** Hiring a vendor to develop a custom solution.
5. **Software-as-a-Service (SaaS):** Subscribing to a cloud-based logistics platform.

Each option had pros and cons:

- In-house development offered customization but required significant time and resources.
- Open source was cost-effective but lacked support and long-term guarantees.
- Off-the-shelf was quick to implement but may not align with all business needs.
- Outsourcing provided expertise but raised concerns about data security and vendor dependency.
- SaaS offered scalability and cost predictability but limited customization and raised data privacy concerns.

Implementation Strategy:

After six months of evaluation, SmartTrans selected a **hybrid approach**:

- A **SaaS-based logistics system** for standard operations (shipment tracking, invoicing).
- A **custom-built analytics module** developed in-house for route optimization and reporting.
- An **API integration layer** to connect with client systems and 3PL providers.

To manage the project, the team adopted an **Agile development methodology** for the custom module and used **Service-Level Agreements (SLAs)** with the SaaS provider to ensure availability and performance metrics.

The implementation was scheduled in three phases over nine months, with parallel staff training and system testing to ensure smooth change management.

Outcomes:

After 12 months, SmartTrans reported the following benefits:

- Shipment delivery times improved by 18%.
- Operating costs reduced by 12% due to better route optimization.
- Customer satisfaction scores increased by 22%.
- IT support tickets dropped significantly due to better user interfaces and system stability.

However, the company faced challenges:

- The initial integration with legacy systems was more complex than anticipated.
- Staff training took longer due to unfamiliar interfaces and mobile apps.
- Managing the vendor relationship required significant legal oversight due to legal and service obligations.

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